



Uncertainty Quantification and Sensitivity Analysis of Spalart Allmaras and Menter Shear Stress Model Coefficients for NACA0012 Airfoil

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This study quantifies epistemic uncertainty associated with the coefficients of two closure models used in Reynolds-Averaged Navier–Stokes simulations. The incompressible flow over a NACA0012 airfoil at angles of attack ranging from 2° to 17° is simulated using the Spalart–Allmaras and Menter’s Shear Stress Transport (SST) turbulence models available in OpenFOAM. For each model, the three most influential coefficients are treated as uncertain parameters. The quantities of interest are the lift and drag coefficients. Uncertainty propagation is performed using non-intrusive polynomial chaos expansion (NIPCE), and global sensitivity analysis is conducted using Sobol indices. The results show that uncertainty increases with the angle of attack, exhibiting a sharp growth near and after stall. Overall, the Menter’s SST model displays higher uncertainty than the Spalart–Allmaras model, and the drag coefficient is more sensitive to model-based uncertainty than the lift coefficient. In the Spalart–Allmaras model, the von Kármán coefficient and σ are the most influential parameter before and post stall, respectively. In contrast, for the Menter’s SST model, the ranking of the most influential parameters on the resulting uncertainty varies with angle of attack.

Nomenclature

C_L	=	Lift coefficient
C_D	=	Drag coefficient
κ	=	von Kármán coefficient
σ	=	Spalart–Allmaras model coefficient
C_{v1}	=	Spalart–Allmaras model coefficient
σ_{ω_1}	=	Menter’s shear stress transport model coefficient
β^*	=	Menter’s shear stress transport model coefficient
a_1	=	Menter’s shear stress transport model coefficient
CI	=	Confidence interval
u_i	=	Mean velocity component in the i -direction
p	=	Pressure
ρ	=	Fluid density
ν	=	Kinematic viscosity
Re	=	Reynolds number
k	=	Turbulent kinetic energy
ϵ	=	Turbulent dissipation rate
ν_t	=	Turbulent kinematic viscosity
ω	=	Specific dissipation rate
S_i	=	Main Sobol index associated with i -th parameter
S_{ij}	=	Dual-interaction Sobol index between i -th and j -th parameters
y^+	=	Wall-normal distance expressed in wall units

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I. Introduction

TURBULENT flows arise in a range of aerospace applications, from flow over wings to flow inside engines, see e.g. [1]. Due to the complex and chaotic nature of turbulent flows, developing a comprehensive physical understanding of turbulent flows remains challenging. Various modelling approaches in Computational Fluid Dynamics (CFD) have been proposed to predict turbulent flows [2]. In particular, the Reynolds-Averaged Navier-Stokes (RANS) approach relies on closure models to represent the effects of turbulence on the mean flow. Numerous closure models have been proposed in the literature, see e.g. [3]. Such models usually contain several parameters or coefficients, the values of which are assigned based on theoretical analyses, empirical correlations, experimental data, and heuristic judgment. The applicability of these coefficients may vary across flow regimes, Reynolds numbers, or geometrical configurations. Consequently, the lack of knowledge regarding the most appropriate coefficient values introduces epistemic uncertainty into RANS predictions of key quantities of interest (QoIs) [4, 5]. It is, therefore, essential to assess the robustness of the turbulence closure models with respect to variations in their coefficients, noting the potential impact on the accuracy and reliability of CFD predictions [5, 6]. This motivates the application of rigorous uncertainty quantification (UQ) techniques to CFD simulations in aerospace engineering.

This study aims to quantify uncertainty and assess sensitivity in aerodynamic QoIs arising from epistemic uncertainty—specifically, uncertainty due to lack of knowledge—in the coefficients of RANS turbulence closure models. We consider two widely used models: the Spalart–Allmaras (SA) model [7] and the Menter’s Shear Stress Transport (SST) model [8]. For each model, the uncertainty due to the three most influential parameters identified by Schaefer et al. [5] are considered, see Section III for details. The QoIs examined are the lift and drag coefficients of a NACA0012 airfoil over angles of attack ranging from 0° to 17° . The flow is assumed to be incompressible and simulated using OpenFOAM [9], a widely-used finite-volume flow solver.

A significant amount of work has been done on UQ of model coefficients for RANS turbulence models under different conditions and using different frameworks. Edeling et al. [10], developed a posterior error estimate of turbulence models using a Bayesian model scenario averaging (BMSA) method. The author, however, concluded that this method was computationally intensive for complex geometries. In previous studies by Schaefer et al. [5, 6], the authors identified the most influential model coefficients for various integrated coefficients and flow QoIs for the SA [7], Menter’s SST [8], and $k - \omega$ [11] turbulence models. The study was carried out for the subsonic flat plate, the RAE 2822 airfoil, and the NASA Common Research Model [12] based on the configuration of the transonic civil aircraft. The studies employed Point Collocation NIPCE for uncertainty propagation and the Sobol Decomposition for global sensitivity analysis. These studies were limited to a fixed Lift coefficient (C_L) or angle of attack. In another study by Schaefer et al. [13], it was shown that these results were independent of the flow solver, the mesh topology, and the scaled dimensionality of the geometry.

In a study by Li et al. [14], Bayesian uncertainty analysis and Bayesian calibration are performed on model coefficients of Menter’s Shear Stress Model [8] with pressure coefficient as the quantity of interest. The study was performed on RAE 2822 airfoil and ONERA M6 wing for transonic flow at three different angles of attack. This study was limited to pressure coefficient as the quantity of interest and limited number of angles of attack. Maruyama et al. [15], quantified the epistemic uncertainty in Spalart Allmaras [7] turbulence model using a Bayesian framework for RAE 2822 airfoil and transonic XRF1 aircraft. The authors concluded that the epistemic uncertainties in closure coefficients result into uncertainties in flow quantities of interest downstream of the shock. Han and Hosder [16], conducted UQ and sensitivity analysis of Spalart Allmaras [7] turbulence model for synthetic jet actuator. The author concluded that the reference experimental velocity profiles were enveloped by the confidence interval of the RANS simulations. In another study by West IV and Hosder [17], the uncertainty propagation in hypersonic re-entry flows was studied employing NIPCE with sparse sampling. The study concluded that accurate surrogate model could be constructed with a limited number of samples.

Yang et al. [18], employed Bayesian optimization to optimally calibrate the coefficients of the Menter’s SST model [8] for wall pressure in supersonic flows. The authors reported a decrease in average relative error through employing optimal model coefficients. Further, the flow field before and after the calibration was consistent, confirming the robustness of the model with respect to the variations in its coefficients. This trend is, however, inconsistent to the study by He et al. [19], which used the metamodel-based Monte Carlo method to investigate the parametric uncertainty of the Spalart Allmaras [7] model for compressor flow features. The author concluded that recalibration of the model coefficients to improve the prediction of specific flow features would lead to degraded accuracy in other QoIs, highlighting the need for modifying the structure of the Spalart Allmaras [7] model. Bin et al. [20], employed Bayesian optimization to recalibrate the model coefficients and functions of Spalart Allmaras [7] model for an axisymmetric jet, airfoil near wake, and a fully developed channel flow. The calibrated model showed improvements in both separated

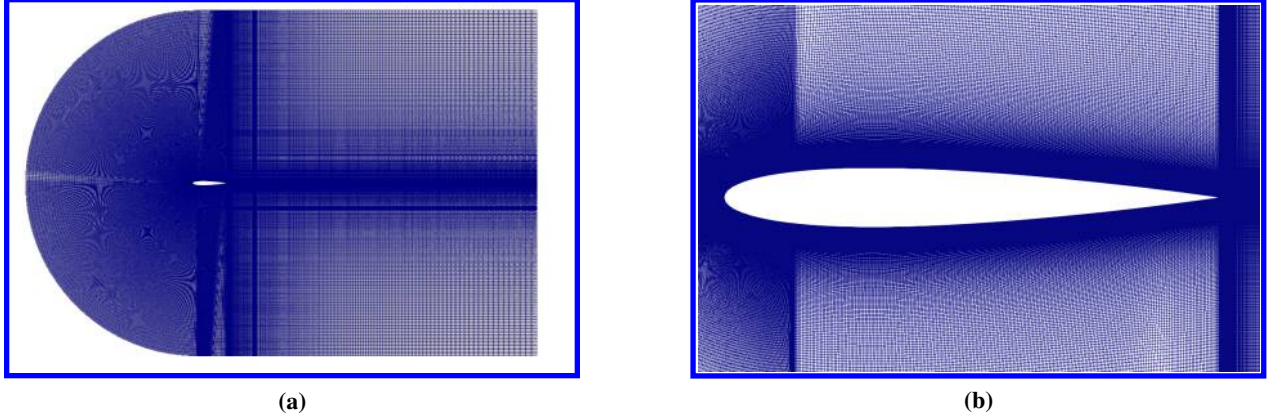


Fig. 1 The structured computational mesh around the NACA0012 airfoil: (a) over the whole domain and (b) zoomed-in around the airfoil.

and equilibrium wall-bounded flows. The improvements achieved by this study were, however, limited by the specific training data used and hence the specific flow configuration. A more comprehensive and structured overview of existing body of research relevant to this topic is provided by Xiao and Cinnella [4].

The remainder of this paper is structured as follows. Section II presents the details of the numerical setup. Section III describes the RANS turbulence models considered in this study. Section IV outlines the uncertainty quantification and sensitivity analysis methodologies. Section V reports and discusses the results. Finally, Section VI provides the conclusions and highlights potential directions for future work.

II. Flow Solver and CFD Case

A. Flow Solver

The flow simulations in this study are performed using OpenFOAM v2012 [9]. OpenFOAM is an open-source, finite-volume-based CFD solver. Steady-state RANS simulations are carried out with the *simpleFoam* solver. A second-order upwind scheme is employed for the discretization of the nonlinear convective terms in the momentum equations, while a central-difference scheme is used for the diffusive terms. To support UQ analysis, an automated Python workflow was developed to handle case setup, systematic modification of the RANS model coefficients, execution of the simulations, and post-processing of the results.

B. CFD Case

The incompressible flow over a two-dimensional NACA0012 airfoil is considered at Reynolds number $Re = 6 \times 10^6$, that is based on the inflow velocity, airfoil chord length, and fluid viscosity. The details of this set-up along with the reference experimental data are taken from the NASA Turbulence Modeling Resource, "2D NACA0012 Validation Case" [21]. The chord length of the airfoil is c and the computational domain considered for meshing is $10c$ to neutralize any blockage affects. A structured C-mesh consisting of 758,000 cells is used in this study, with refined resolution near the wall to maintain $y^+ < 1$ along the airfoil surface in all the cases, see Figure 1. This mesh resolution was consolidated after conducting a mesh convergence study at 2° angle of attack with Menter's SST model, see Figure 2. The angle of attack was varied between 2° to 14° at an increment of 2 and an additional 17° was considered in the stall zone. The angle of attack was varied by changing the direction of freestream velocity vector in the inflow condition, while keeping the geometry and mesh fixed.

III. Turbulence Models

This section briefly describes the RANS models used in this study along with their main equations and uncertain parameters. For the latter, information regarding the default values and variation ranges in the UQ analyses is provided.

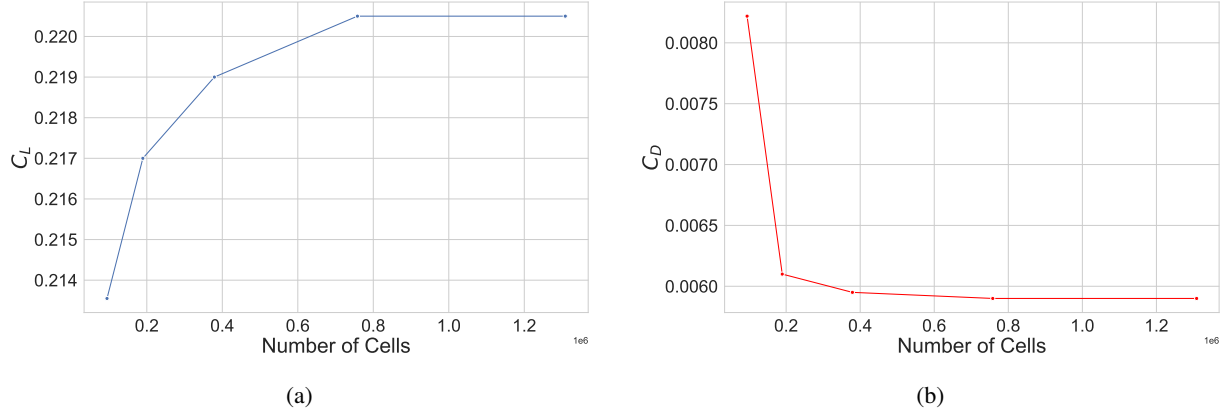


Fig. 2 Mesh convergence study at 2° angle of attack using the Menter's SST model: variation of the (a) lift and (b) drag coefficients with the number of mesh cells.

A. The Spalart-Allmaras Model

The Spalart-Allmaras (SA) model [7] is a one-equation eddy viscosity-based turbulence model developed to efficiently simulate wall-bounded turbulent flows, particularly in aerospace applications. This model is an evolved model of Nee-Kovansky model [22] and is inspired by Baldwin-Barth model [23]. It was designed for high Reynolds-number flows, providing a balance between accuracy and computational efficiency. The model has been widely used in RANS simulations of external aerodynamic flows, including flow over airfoils, see e.g. [5]. However, it has limitations, including its reduced accuracy in low Reynolds-number flows, inability to model flow transition and separation effectively [3]. The differential equation for the scaled eddy kinematic viscosity $\tilde{\nu}$ is written as:

$$\frac{\partial \tilde{\nu}}{\partial t} + u_j \frac{\partial \tilde{\nu}}{\partial x_j} = C_{b1}(1 - f_{t2})\tilde{S}\tilde{\nu} + \frac{1}{\sigma} \left[\frac{\partial}{\partial x_j} \left((\nu + \tilde{\nu}) \frac{\partial \tilde{\nu}}{\partial x_j} \right) + C_{b2} \frac{\partial \tilde{\nu}}{\partial x_j} \frac{\partial \tilde{\nu}}{\partial x_j} \right] - C_{w1} f_w \left(\frac{\tilde{\nu}}{d} \right)^2 \quad (1)$$

where, t is time, x_j refers to the spatial coordinate in the j -th direction, ν denotes the kinematic viscosity, and d is the distance of a point in the flow domain from the closest wall surface. The eddy viscosity ν_t and model coefficients are given by the following:

$$\nu_t = \tilde{\nu} f_{v1} \quad (2)$$

$$f_{v1} = \frac{\chi^3}{\chi^3 + C_{v1}^3}, \quad \chi = \frac{\tilde{\nu}}{\nu} \quad (3)$$

$$\tilde{S} = S + \frac{\tilde{\nu}}{\kappa^2 d^2} f_{v2}, \quad f_{v2} = 1 - \frac{\chi}{1 + \chi f_{v1}} \quad (4)$$

$$f_w = g \left[\frac{1 + C_{w3}^6}{g^6 + C_{w3}^6} \right]^{1/6}, \quad g = r + C_{w2}(r^6 - r), \quad r = \frac{\tilde{\nu}}{\tilde{S} \kappa^2 d^2} \quad (5)$$

The standard value of the coefficients along with their associated variation range are summarized in Table 1. These are taken from Schaefer et al. [5, 6] and Spalart and Allmaras [7]. The three most significant model coefficients contributing to uncertainty identified by Schaefer et al. [5, 6] have been selected for the UQ analyses in this study. These are σ , κ , and C_{v1} with admissible variation ranges shown in Table 1.

B. The Menter's Shear Stress Transport Model

The Menter's SST Model [8] is a two-equation model which is a blend of the $k - \omega$ [11] and the $k - \epsilon$ [24] RANS models. The model uses a blending function to switch between the $k - \omega$ model [3] in the near-wall region and the $k - \epsilon$ model [24] in the outer-layer of the boundary layer to enhance the accuracy in a wide range of turbulent flows. The model was designed to improve prediction of the flow separation under adverse pressure-gradient, particularly

Coefficient	Standard Value	Lower Bound	Upper Bound
σ	2/3	0.6	1.0
κ	0.41	0.38	0.42
c_{v1}	7.1	6.9	7.3
c_{w3}	2.0	1.75	2.5
$ct1$	1.0	0.1	10.0
$ct2$	2.0	1.9	2.1
c_{b1}	0.1355	0.12893	0.13700
c_{b2}	0.622	0.60983	0.68750
c_{w2}	0.3	0.05500	0.35250

Table 1 The coefficients in the SA model: their standard values and variation range [5–7].

Coefficient	Standard Value	Lower Bound	Upper Bound
σ_{k1}	0.85	0.7	1.0
σ_{k2}	1.0	0.8	1.2
$\sigma_{\omega 1}$	0.5	0.3	0.7
$\sigma_{\omega 2}$	0.856	0.7	1.0
β^*/β_1	1.20	1.19	1.31
β^*/β_2	1.0870	1.05	1.45
β^*	0.09	0.0784	0.1024
κ	0.41	0.38	0.42
a_1	0.31	0.31	0.40

Table 2 The coefficients in the Menter’s SST model: their standard values and variation range [6, 8].

in aerospace and turbomachinery flows. The SST model, however, assumes isotropic turbulence that may introduce limitations in complex anisotropic flows. The governing equations of the SST model describe the evolution of the turbulent kinetic energy k and specific dissipation rate ω , and are given by:

$$\frac{\partial k}{\partial t} + u_j \frac{\partial k}{\partial x_j} = P_k - \beta^* k \omega + \frac{\partial}{\partial x_j} \left[(\nu + \sigma_k \nu_t) \frac{\partial k}{\partial x_j} \right], \quad (6)$$

$$\frac{\partial \omega}{\partial t} + u_j \frac{\partial \omega}{\partial x_j} = \frac{\omega}{k} P_k - \beta \omega^2 + \frac{\partial}{\partial x_j} \left[(\nu + \sigma_\omega \nu_t) \frac{\partial \omega}{\partial x_j} \right] + 2(1 - F_1) \frac{\sigma_{\omega 2}}{\omega} \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j}, \quad (7)$$

where P_k is the turbulent kinetic energy production and u_j is the averaged velocity component in the j -th direction. The model coefficients are blended using the function F_1 , which is to impose the correct behavior in each flow region. The default values of the SST model’s coefficients and their variation ranges (used by Schaefer et al. [6]) are summarized in Table 2. In this study, we consider three uncertain parameters β^* , $\sigma_{\omega 2}$, and a_1 .

IV. Uncertainty Quantification Approach

In general, there are two types of uncertainties: aleatory and epistemic [25]. The aleatory uncertainties are random, inherent to a system, and usually cannot be fully removed. Examples are the fluctuations in the ambient temperature and pressure when running a wind tunnel experiment. In contrast, the epistemic (systematic) uncertainties originate from a lack of knowledge or modelling insufficiency and discrepancy. Therefore, they can be reduced by improving our knowledge or modelling approach. In the context of the present study, the uncertainty due to the RANS models’ coefficients are epistemic. The general approach in UQ is to express such a problem in a stochastic framework to apply statistical or probabilistic techniques. This means, we let the uncertain model coefficients to vary over prescribed ranges, and then we quantify the resulting uncertainty propagated into the QoIs (here the lift and drag coefficients). As a

complement to this, global sensitivity analyses in the form of the Sobol indices [26] allow for quantifying the share of each of the uncertain parameters in the resulting uncertainty in the QoIs. This section provides a brief description of the methods employed for these purposes. It is noteworthy that all UQ analyses in this study are performed using the open-source Python package UQit [27].

A. Polynomial Chaos Expansion Method

In this study, non-intrusive polynomial chaos expansion (NIPCE) method, see Ghanem and Spanos [28] and Eldred [29], is employed for efficient uncertainty propagation from a limited number of realizations. The non-intrusive nature of the method means that, first, a set of n samples for the uncertain parameters $\mathbf{q}$ are drawn from their admissible space $\mathcal{Q} \subset \mathbb{R}^p$. Then, the flow simulations are performed at each of such samples to obtain the associated values of the QoIs. As thoroughly explained in Rezaeiravesh et al. [30], using the collected data, the NIPCE method constructs a projection-based surrogate for the QoIs in the space of uncertain parameters $\mathbf{q}$. The resulting surrogate reads as,

$$\tilde{f}(\mathbf{q}) = \sum_{k=0}^K \hat{f}_k \Psi_k(\boldsymbol{\xi}), \quad (8)$$

where $\Psi_k(\boldsymbol{\xi}) = \prod_{i=1}^p \psi_{k_i}(\xi_i)$ represents the multivariate basis function and ξ_i corresponds the i -th standardized uncertain parameter ($i = 1, 2, \dots, p$). The single-variate orthogonal basis functions $\psi_{k_i}(\xi_i)$ are chosen based on the generalized PCE rule [31]. In this study, all uncertain parameters are assumed to have a uniform distribution, the Legendre polynomials are used as the basis functions. For each of the uncertain parameters, we draw 4 samples that are the Gauss-Legendre quadrature points [32]. Adopting a tensor-product rule, the total number of samples is 4^3 for $p = 3$ (three uncertain parameters) that results in $K = 63$. Unlike other methods like Monte Carlo, the NIPCE approach ensures computationally efficient uncertainty propagation by keeping the number of realization at minimum. Once the PCE surrogate is constructed, the stochastic mean and variance of $f(\mathbf{q})$ can be directly computed. The estimated variance is used to construct a 95% confidence interval around the stochastic mean.

B. Global Sensitivity Analysis

In most engineering applications, local sensitivity analysis is used to assess the impact of small perturbations in input parameters on the nominal model output. However, this approach relies on linearization and does not account for the full distribution of input uncertainties [25]. In contrast, global sensitivity analysis (GSA) quantifies the variability in model outputs arising from uncertainties in all input parameters simultaneously, capturing their interactions without requiring linearization. Therefore, the GSA provides a comprehensive view of the influence of each parameter on the output's uncertainty. Here, we use the Sobol sensitivity indices [26] to measure global sensitivity. The Sobol decomposition of a model function $f(\mathbf{q})$, up to dual interactions, is written as,

$$f(\mathbf{q}) = f_0 + \sum_{i=1}^p f_i(q_i) + \sum_{1 \leq i < j \leq p} f_{ij}(q_i, q_j), \quad (9)$$

where f_0 is the stochastic mean of $f(\mathbf{q})$, and f_i and f_{ij} denote the main and dual interaction effects, respectively. Similarly, the stochastic variance of $f(\mathbf{q})$ can be decomposed as,

$$\mathbb{V}_{\mathbf{q}}[f(\mathbf{q})] = D = \sum_{i=1}^p D_i + \sum_{1 \leq i < j \leq p} D_{ij} \quad (10)$$

The main and dual-interaction Sobol indices are the contribution of each D_i and D_{ij} , respectively, to the total variance D :

$$S_i = \frac{D_i}{D}, \quad S_{ij} = \frac{D_{ij}}{D}, \quad i, j = 1, \dots, p \quad (11)$$

These satisfy the following expression:

$$\sum_{i=1}^p S_i + \sum_{1 \leq i < j \leq p} S_{ij} = 1 \quad (12)$$

The total number of distinct main and dual-interaction Sobol indices can be calculated via $p + p(p-1)/2$.

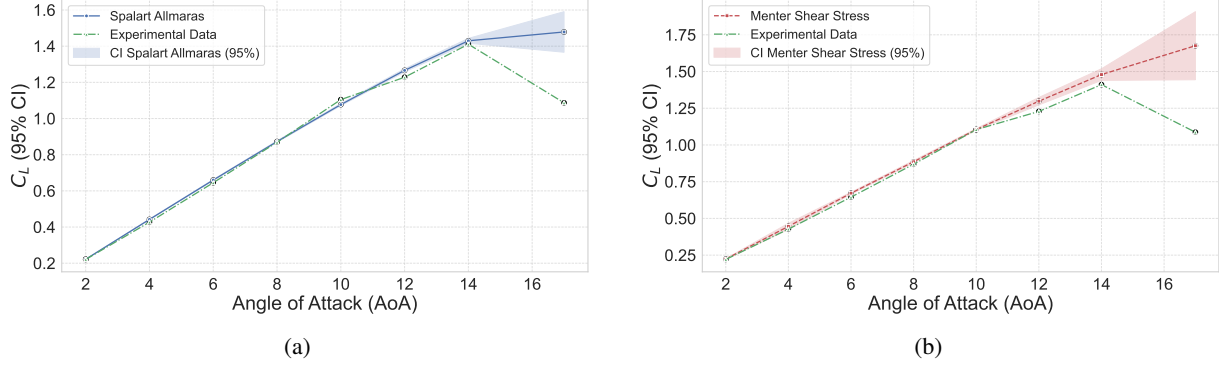


Fig. 3 Variation of the lift coefficient C_L with the angle of attack using (a) the SA and (b) Menter's SST models. The 95% confidence intervals are shown as shaded areas around the stochastic mean predictions. The experimental data are taken from [21].

V. Results and Discussion

A. Uncertainty Quantification

First, the propagated uncertainty from the SA and SST models' parameters into the computed lift and drag coefficients, C_L and C_D , respectively, is discussed. Figure 3 shows the variation of the C_L with the angle of attack (AoA) for both SA and SST RANS models. In the plots, the shaded areas show the 95% Confidence Interval (CI) around the stochastic mean of the QoIs, obtained from the NIPCE. In addition, the experimental data [21] are added for validation of the RANS predictions. For both the turbulence models, the uncertainty in C_L is insignificant up to AoA=10° and then gradually increases up to 14°. Above this, a sharp increase in the 95% CI is observed due to the stall. This means that, in the stall condition, both models become too sensitive to the choice of the coefficients. Nevertheless, no combination of the modelling coefficients could result in accurate prediction of the C_L at large AoAs where separation occurs. This is clear from the deviation between the mean prediction and the reference experimental data.

A similar trend is observed in the uncertainty of C_D values, see Figure 4. The uncertainty in C_D is low up to AoA of 12° and it significantly grows in the stall region. What is important is that, in general, the confidence interval for the SA model is smaller than the SST model indicating more robustness of the SA model to the variation of its coefficients. In contrast, for the SST model, the reference experimental data fall within the CI at all AoAs, while for the SA model, this only happens for AoAs from 8° to 12°. This behaviour should be interpreted cautiously noting the potential uncertainty in the experimental data is not included. Another important observation is that the sensitivity of the C_D to the variation of the modelling coefficients is more than that of the C_L for both RANS models and across all AoAs. This can be better understood from Figure 5, where the normalized uncertainty ($1.96 \times \text{standard deviation} / \text{mean}$) in both QoIs is plotted at different AoAs. Clearly, at any AoA the uncertainty in the C_D is much higher than that in the C_L . For both QoIs, there is a sharp increase in the uncertainty after the stall. What is also more evident is the higher uncertainty in the QoIs when using the SST model. When interpreting this, we should also note that the uncertain parameters and their variation range were not the same in the SA and SST models.

B. Sensitivity Analysis

The Sobol indices can quantify the contribution of the uncertain parameters to the propagated uncertainty in the QoIs. Figure 6 shows the main and dual-interaction Sobol indices for the C_L and C_D when the SA model is adopted. Looking at the main indices, it is clear that for all AoAs except the highest one considered in this study, the von Kármán coefficient κ is the most influential parameter on both C_L and C_D . This is only in the stall region, AoA greater than 14° that the sensitivity with respect to κ decreases while seeing an increase with respect to σ . These observations are consistent with what is reported by Schaefer et al., [5, 6]. Interestingly, the sensitivity with respect to C_{v_1} remains almost constant at all AoAs, although this parameter ranks second and third, in terms of influence, on the C_L and C_D , respectively. A main interpretation from these observations is that if there is only one parameter that can be altered to control the lift and drag coefficients, it will be κ . In contrast to the main indices, the dual interactions between the three parameters κ , σ and C_{v_1} are observed to be negligible. It is only after the stall, that the interaction Sobol indices increase although

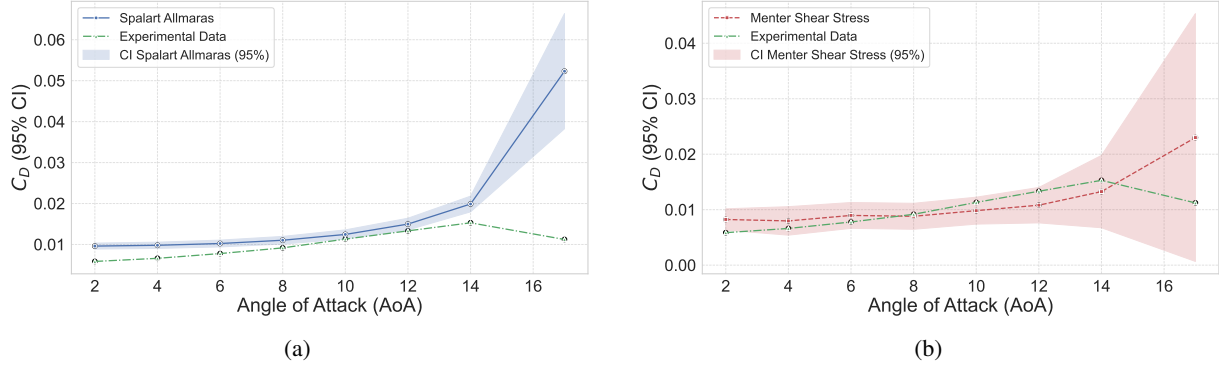


Fig. 4 Variation of the drag coefficient C_D with the angle of attack using (a) the SA and (b) Menter's SST models. The 95% confidence intervals are shown as shaded areas around the stochastic mean predictions. The experimental data are taken from [21].

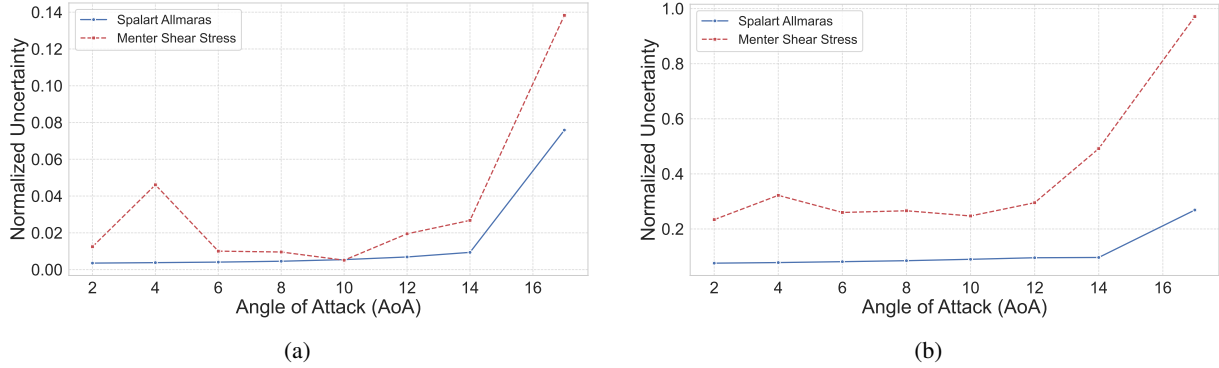


Fig. 5 Variation of the normalized uncertainty in the (a) C_L and (b) C_D resulted from uncertainty in the selected parameters of the SA and SST models.

they are still insignificant compared to the main ones. Relatively, the sensitivity of the C_L and C_D to the dual interaction between $\kappa - \sigma$ is observed to be larger than the other two interactions.

Figure 7 show the Sobol Indices for Menter's SST [8] model. Compared to the SA model (Figure 6), a less consistent variation of the Sobol indices across AoAs is observed. Particularly for C_L , the values of the main Sobol indices with respect to the parameters σ_{ω_1} , β^* , and a_1 are comparable to each other. At lower AoAs, there is a lack of dominant influential parameter. In contrast, for the drag coefficient, the parameter β^* shows the highest influence at the AoA below 14° . Further, σ_{ω_1} and a_1 consistently rank as the second and third most influential parameters, respectively. Furthermore, the parameter a_1 contributes most to the uncertainty in the stall region with an increasing influence with AoA larger than 14° .

In contrast to the SA model, the dual interactions between the uncertain parameters of the SST model are generally larger and often comparable to the associated main indices. For C_L , the variation of the dual interactions with AoA is less consistent, although a large peak for $\sigma_{\omega_1} - \beta^*$ is observed at the 10° AoA. This interaction Sobol index is larger than all the main indices, indicating that around this specific AoA, the simultaneous variation in σ_{ω_1} and β^* is the factor that drives the overall uncertainty in C_L . For C_D , however, the variation of the dual interaction indices across different values of the AoA exhibits a more consistent behavior. For AoA smaller than 12° , the dual interactions are negligible. Above 12° , the index measuring the sensitivity with respect to $\beta^* - a_1$ consistently increases and becomes the largest among all Sobol indices in the stall region. It is important to note that in the same range of AoA, the dominant main Sobol index, namely that with respect to β^* , gradually decreases. The overall conclusion for the SST model is that, for controlling the accuracy of C_D , the variations in β^* and $\beta^* - a_1$ are the most influential factors before and after stall, respectively. However, any such conclusion for C_L is not suggested from the analysis.

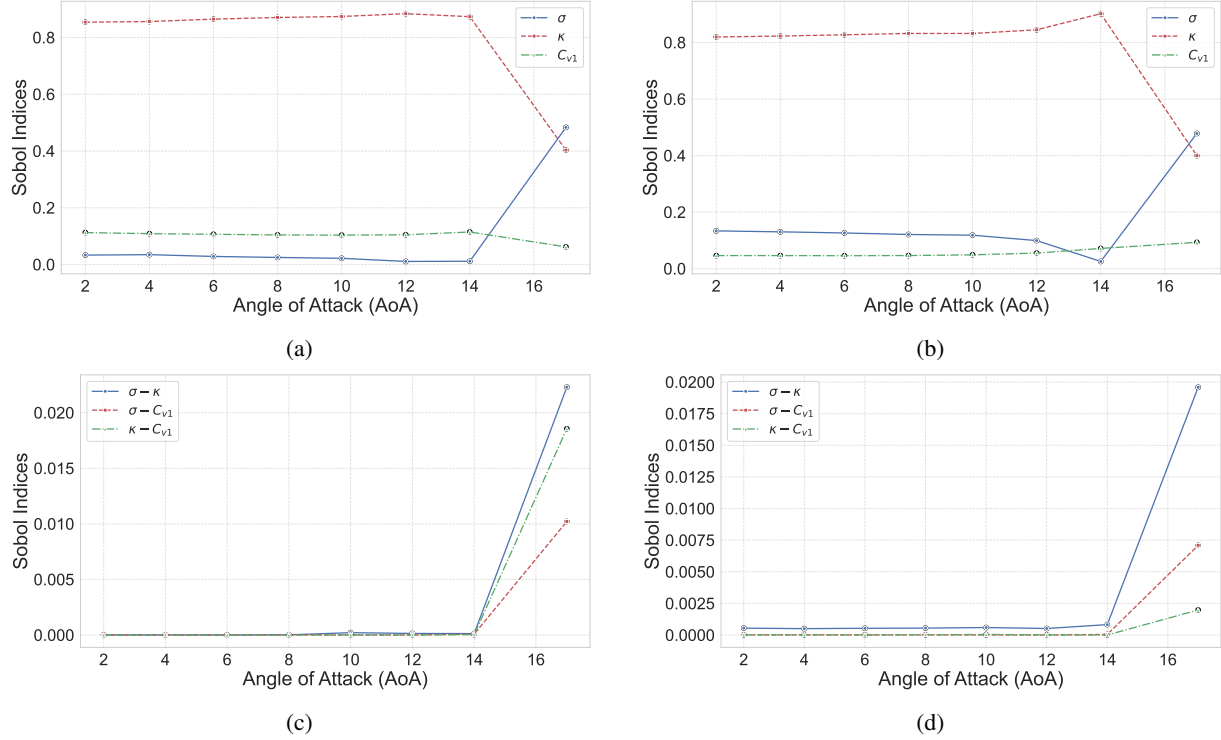


Fig. 6 Main Sobol sensitivity indices of (a) C_L and (b) C_D , as well as the dual-interaction Sobol indices of (c) C_L and (d) C_D with respect to variations in the uncertain parameters σ , κ , and C_{v1} in the SA model.

VI. Conclusion and Future Work

A. Conclusion

This study conducted uncertainty quantification and global sensitivity analysis for two RANS turbulence models. The uncertainties were introduced through three modelling parameters in each model. The quantities of interest were the lift and drag coefficients of the NACA0012 airfoil in incompressible flow, computed from CFD simulations over a range of angles of attack. The uncertainty propagation results showed that the uncertainty increases with angle of attack, with a sharp rise near the stall. In general, Menter's SST model [8] exhibited higher uncertainty (lower robustness) compared to the Spalart-Allmaras (SA) model [7]. Overall, the drag coefficient C_D showed a higher sensitivity to variations in the RANS model parameters than the lift coefficient C_L .

The global sensitivity analysis, based on Sobol indices, showed that up to the stall the von Kármán coefficient κ is the most influential parameter driving the uncertainty in both C_L and C_D when using the SA model. After stall, the influence of κ and another parameter becomes nearly identical. For the SA model, interactions between model coefficients were found to be insignificant at all angles of attack except beyond stall. For the SST model, no single parameter consistently dominated the uncertainty in C_L across the angles of attack. In contrast, the parameter β^* was the most influential factor for C_D before stall, while after stall the interaction between β^* and a_1 became the dominant contributor to uncertainty in C_D . Overall, this study provides a systematic uncertainty quantification framework for comparing RANS models when the influence of their parameters is of interest. Moreover, for the NACA0012 case considered here, the analyses suggest that the SA model is more robust and easier to control than the SST model.

B. Future Work

This study was limited to two RANS models for the NACA0012 airfoil in incompressible flow regime and a fixed Reynolds number. Future studies should focus on uncertainty quantification for multiple airfoils at different angles of attack to see if similar trends emerge or if a slight change in geometry, as seen in different airfoils, change the sensitivity trends. In addition to this, there is a need to analyze how the trends for epistemic uncertainty quantification and

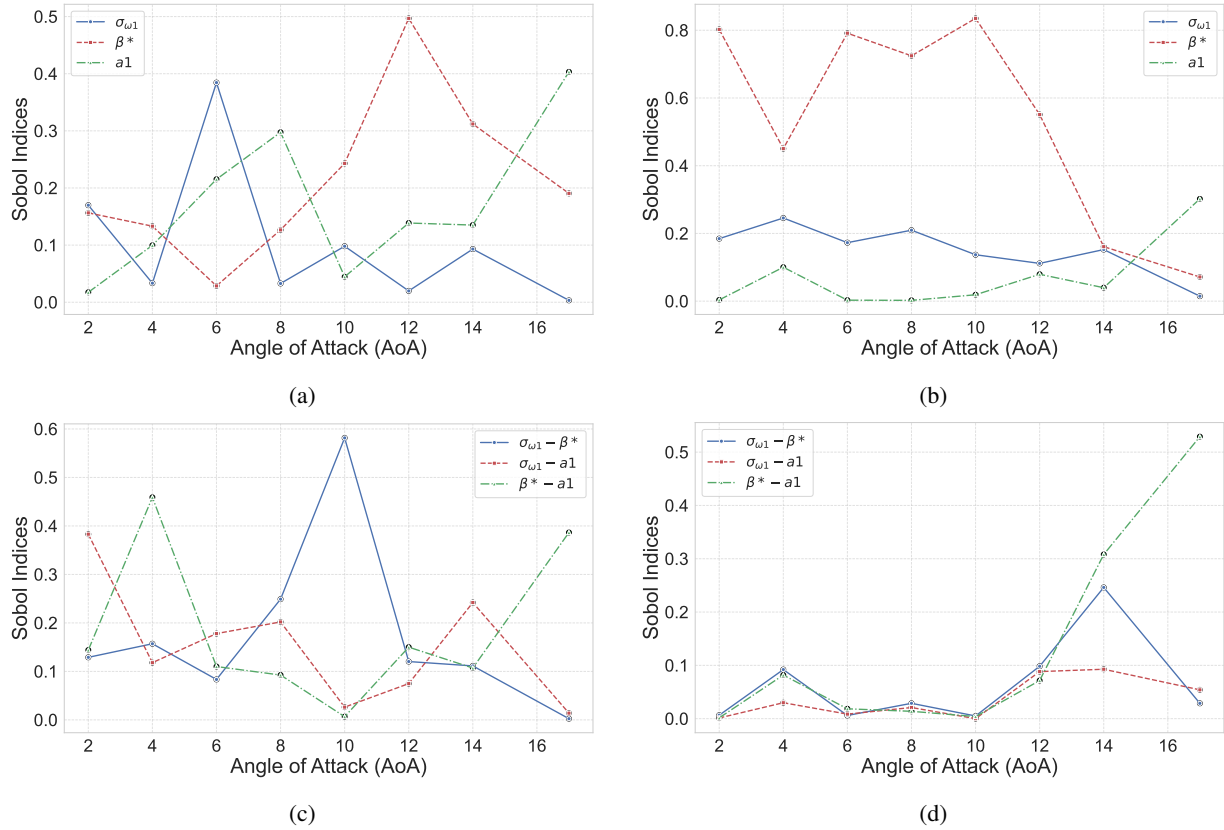


Fig. 7 Main Sobol sensitivity indices of (a) C_L and (b) C_D , as well as the dual-interaction Sobol indices of (c) C_L and (d) C_D with respect to variations in the uncertain parameters $\sigma_{\omega 1}$, β^* , and a_1 in the SST model.

sensitivity analysis emerge for different flow regimes of subsonic compressible, transonic, supersonic and hypersonic flows. These analyses can also be extended to three-dimensional wings and complex geometries with variations in geometry, although the computational cost of multiple CFD simulations may become a concern.

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